

## PALESTINE, TX, USA

WMO: 720314

Lat: 31.780N

Lon: 95.706W

Elev: 423

StdP: 14.47

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 93983

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 26.4       | 29.6 | 12.4                          | 10.5 | 37.8 | 17.6 | 13.6 | 40.4 | 20.9                  | 52.1 | 18.9 | 54.5 | 3.9                   | 340  | 0.389 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 20.8                      | 99.6            | 73.6 | 97.4 | 74.0 | 95.3 | 74.4 | 79.2                | 89.0 | 78.4 | 88.0 | 77.8 | 87.2 | 6.3                     | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.1                            | 143.3 | 82.3 | 76.3 | 139.2 | 81.8 | 75.4 | 135.1 | 81.4 | 43.0          | 89.3 | 42.2 | 88.3 | 41.5 | 87.1 | 82.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |     |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-----|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |     |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p) |       |
| 17.6              | 15.7 | 12.9 | DB                         | 18.6 | 102.8              | 4.5 | 2.7                                                | 15.4 | 104.7      | 12.8 | 106.3      | 10.3 | 107.8      | 7.0 | 109.8 |
|                   |      |      | WB                         | 17.6 | 80.0               | 4.0 | 1.2                                                | 14.7 | 80.9       | 12.3 | 81.6       | 10.1 | 82.3       | 7.1 | 83.1  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 67.4   | 48.8  | 52.7  | 60.1  | 66.1  | 73.7  | 81.0  | 83.6  | 84.6  | 79.3  | 69.0  | 58.3  | 50.8 |
|                                                                                  | DBStd       | 14.81  | 9.94  | 10.69 | 9.50  | 7.80  | 6.31  | 3.97  | 3.33  | 3.82  | 5.10  | 8.21  | 9.64  | 9.83 |
|                                                                                  | HDD50       | 410    | 143   | 89    | 27    | 3     | 0     | 0     | 0     | 0     | 0     | 1     | 34    | 113  |
|                                                                                  | HDD65       | 1923   | 511   | 362   | 206   | 80    | 12    | 0     | 0     | 0     | 1     | 55    | 242   | 454  |
|                                                                                  | CDD50       | 6763   | 105   | 164   | 341   | 487   | 736   | 929   | 1042  | 1073  | 878   | 591   | 281   | 136  |
|                                                                                  | CDD65       | 2801   | 8     | 17    | 55    | 114   | 283   | 479   | 577   | 608   | 429   | 180   | 39    | 12   |
|                                                                                  | CDH74       | 29652  | 32    | 83    | 317   | 875   | 2542  | 5241  | 6785  | 7550  | 4410  | 1568  | 205   | 44   |
|                                                                                  | CDH80       | 13947  | 1     | 6     | 41    | 191   | 910   | 2525  | 3506  | 4124  | 2088  | 526   | 27    | 2    |
| Wind                                                                             | WSAvg       | 5.7    | 6.3   | 6.8   | 7.1   | 6.9   | 6.0   | 5.1   | 4.6   | 4.5   | 4.4   | 5.1   | 5.6   | 5.8  |
| Precipitation                                                                    | PrecAvg     | 46.7   | 4.1   | 4.0   | 3.8   | 3.7   | 3.9   | 4.3   | 2.7   | 3.3   | 3.7   | 4.5   | 3.8   | 4.2  |
|                                                                                  | PrecMax     | 65.4   | 8.2   | 9.0   | 7.1   | 8.7   | 7.8   | 9.9   | 10.8  | 9.6   | 7.6   | 12.4  | 12.6  | 9.1  |
|                                                                                  | PrecMin     | 26.6   | 0.4   | 0.1   | 0.9   | 0.7   | 0.0   | 0.0   | 0.0   | 0.0   | 1.5   | 0.0   | 0.2   | 0.6  |
|                                                                                  | PrecStd     | 10.6   | 2.2   | 2.3   | 1.7   | 2.0   | 2.4   | 2.7   | 2.3   | 2.7   | 1.8   | 3.4   | 2.7   | 2.4  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 77.5  | 81.0  | 83.9  | 89.9  | 93.4  | 99.8  | 101.4 | 104.5 | 99.8  | 91.4  | 83.2  | 78.8 |
|                                                                                  |             | MCWB   | 62.6  | 63.0  | 65.6  | 71.3  | 73.4  | 73.0  | 73.0  | 73.2  | 70.3  | 71.9  | 70.0  | 65.5 |
|                                                                                  | 2%          | DB     | 73.2  | 76.8  | 80.9  | 84.7  | 90.7  | 96.4  | 99.0  | 100.1 | 96.5  | 88.7  | 79.4  | 74.3 |
|                                                                                  |             | MCWB   | 61.7  | 63.1  | 65.8  | 68.4  | 72.9  | 73.2  | 73.4  | 74.5  | 72.5  | 71.6  | 67.1  | 65.0 |
|                                                                                  | 5%          | DB     | 70.1  | 72.7  | 77.5  | 81.6  | 88.0  | 93.5  | 96.7  | 98.5  | 93.0  | 86.0  | 76.0  | 70.9 |
|                                                                                  |             | MCWB   | 61.6  | 62.7  | 64.6  | 67.4  | 72.6  | 74.3  | 73.9  | 74.8  | 73.4  | 70.7  | 65.8  | 63.9 |
|                                                                                  | 10%         | DB     | 66.0  | 69.7  | 74.4  | 79.2  | 85.6  | 91.3  | 94.1  | 95.8  | 90.6  | 82.3  | 72.8  | 68.0 |
|                                                                                  |             | MCWB   | 58.7  | 61.7  | 62.7  | 66.6  | 71.9  | 74.9  | 75.4  | 75.0  | 73.0  | 68.9  | 65.5  | 61.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 67.8  | 69.3  | 70.5  | 74.7  | 77.6  | 80.1  | 80.1  | 80.3  | 79.1  | 77.5  | 72.9  | 70.5 |
|                                                                                  |             | MCDB   | 73.0  | 74.0  | 77.5  | 81.7  | 85.4  | 89.3  | 89.4  | 89.3  | 88.2  | 84.1  | 78.3  | 73.6 |
|                                                                                  | 2%          | WB     | 65.9  | 67.5  | 68.9  | 72.3  | 76.0  | 78.7  | 79.1  | 79.1  | 77.9  | 75.7  | 70.9  | 67.7 |
|                                                                                  |             | MCDB   | 71.1  | 71.9  | 76.0  | 80.4  | 84.5  | 87.7  | 88.8  | 89.5  | 86.8  | 82.9  | 75.9  | 72.1 |
|                                                                                  | 5%          | WB     | 63.7  | 65.3  | 67.3  | 70.9  | 74.8  | 77.8  | 78.3  | 78.3  | 77.0  | 74.5  | 69.1  | 65.6 |
|                                                                                  |             | MCDB   | 68.0  | 70.7  | 74.6  | 78.4  | 83.5  | 86.5  | 87.8  | 88.5  | 85.6  | 81.1  | 73.7  | 70.5 |
|                                                                                  | 10%         | WB     | 59.6  | 62.9  | 65.8  | 69.2  | 73.7  | 76.8  | 77.5  | 77.6  | 76.1  | 73.0  | 66.7  | 62.9 |
|                                                                                  |             | MCDB   | 64.5  | 68.3  | 72.3  | 76.1  | 82.2  | 85.3  | 86.8  | 87.7  | 84.2  | 79.1  | 72.0  | 66.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 20.5  | 19.5  | 20.5  | 21.7  | 19.9  | 19.1  | 20.0  | 20.8  | 20.6  | 21.8  | 20.8  | 19.7 |
|                                                                                  |             | MCDBR  | 23.6  | 21.7  | 20.8  | 21.9  | 21.6  | 22.0  | 23.6  | 24.3  | 23.4  | 22.3  | 21.1  | 21.6 |
|                                                                                  |             | MCWBR  | 13.7  | 12.8  | 10.2  | 9.9   | 7.8   | 5.8   | 5.1   | 5.4   | 6.7   | 8.5   | 11.5  | 14.4 |
|                                                                                  | 5% WB       | MCDBR  | 18.9  | 18.2  | 18.0  | 18.1  | 18.3  | 17.8  | 18.4  | 19.9  | 18.5  | 17.8  | 17.9  | 18.4 |
|                                                                                  | MCWBR       | 14.4   | 12.6  | 11.0  | 10.0  | 8.1   | 6.2   | 5.6   | 5.6   | 6.2   | 8.3   | 12.1  | 14.6  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.320  | 0.341 | 0.363 | 0.405 | 0.440 | 0.449 | 0.456 | 0.458 | 0.416 | 0.364 | 0.353 | 0.327 |      |
|                                                                                  | taud        | 2.512  | 2.464 | 2.408 | 2.303 | 2.258 | 2.275 | 2.289 | 2.275 | 2.346 | 2.471 | 2.451 | 2.500 |      |
|                                                                                  | Ebn at Noon | 288    | 291   | 291   | 281   | 271   | 267   | 265   | 263   | 271   | 279   | 272   | 278   |      |
|                                                                                  | Edh at Noon | 28     | 32    | 36    | 42    | 44    | 43    | 42    | 42    | 38    | 31    | 29    | 27    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 893    | 1004  | 1375  | 1704  | 1838  | 2022  | 2057  | 1926  | 1625  | 1314  | 959   | 780   |      |
|                                                                                  | RadStd      | 84     | 177   | 128   | 115   | 138   | 170   | 143   | 136   | 143   | 184   | 119   | 102   |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (40 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +203  | +158  |

Nomenclature: See separate page